Karthik Patel

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PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienced in designing fault-tolerant microservices, optimizing database throughput, and shipping production code with rigorous test coverage.

EDUCATION

B.E. in Computer Science — Delhi Technological University *2021 - 2025*

Cumulative Grade Point Average: 9.60/10.0

WORK EXPERIENCE

Software Engineer - Core Services — Razorpay Software Pvt. Ltd., Bengaluru, India *Jun 2024 - Present*

- Migrated monolithic checkout service to gRPC microservices, shrinking cloud compute costs by 32% and reducing deployment cycle times from 4 hours to 12 minutes.
- Architected high-through-put payment webhook processor using Go and Redis Streams, handling 14,000 requests/sec with p99 latency under 28ms.
- Developed responsive internal operations dashboard in React and TypeScript with TanStack Query, accelerating merchant dispute resolution by 40%.

TECHNICAL PROJECTS

Peer-to-Peer File Transfer Protocol with NAT Traversal | Technologies: Rust, STUN/TURN, UDP, Tokio, Cryptography

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.

Distributed Log-Structured Key-Value Store | Technologies: Go, Raft Consensus, gRPC, LSM-Tree, Docker

- Implemented distributed LSM-tree storage engine in Go with write-ahead logging (WAL), SSTable compaction, and bloom filters.
- Embedded Raft consensus algorithm for distributed multi-node replication, surviving arbitrary node crashes without state divergence.
- Benchmarked performance using custom concurrent load generator, achieving 85,000 read operations/sec across a 5-node cluster.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines